

Jose Gabriel Mercado Vasquez

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Portfolio: <https://gabrielmervac.github.io>

EDUCATION

National Autonomous University of Mexico , Mexico City, Mexico Ph.D., Physics Thesis: "Random Searches in Intermittent Environments"	2022
National Autonomous University of Mexico , Mexico City, Mexico M. Sc., Physics Thesis: "Lotka--Volterra systems with stochastic resetting"	2018
National Autonomous University of Mexico , Mexico City, Mexico B.S., Physics Thesis: "Automatization of the Z-scan technique and its application to the non-linear refractive index and the third order absorption coefficient for PMMA films doped with DR1"	2015

AREAS OF EXPERTISE

Mathematical modelling; Computational simulation; Programming; Machine learning; Deep learning; Stochastic processes; Normal and anomalous diffusion; Cell signaling; Immune response.

RESEARCH EXPERIENCE

The University of Chicago , Pritzker School of Molecular Engineering <i>Postdoctoral Researcher, Laboratory of Prof. Savas Tay</i> • Combined microfluidic technology with fluorescent microscopy to study the response of immune cells to transient perturbations • Developed mathematical models to understand the dynamics of the immune response of cells in an infectious disease and cell to cell communication • Leveraged machine learning and deep learning models to predict the evolution of immune response • Collaborated with researchers from different disciplines in biology, chemistry, material science, and physics to create a microfluidic platform for high-throughput antibiotic screening	2022-Present
National Autonomous University of Mexico , Institute of Physics <i>Doctoral Researcher, Laboratory of Prof. Denis Boyer</i> • Studied the statistical properties of the first passage time of random searches in the presence of fluctuating environments • Collaborated with researchers from international institutions to generate a physical model that implements stochastic resetting to optimize random searches • Investigated the evolution of population dynamics in a predator-prey system by integrating spatial mobility	2018-2022
University Paris-Saclay , Laboratory of Theoretical Physics and Statistical Models <i>Summer Graduate Researcher, Laboratory of Prof. Satya Majumdar</i> • Studied the distribution of the arrival times to an absorbing target by a finite-lived searcher	2022

- Automated an optical setup to measure the nonlinear optical properties of organic materials

PUBLICATIONS (* denotes equal contribution)

- 1) **Gabriel Mercado-Vásquez** and Denis Boyer. (2026) "Stochastic synthesis-degradation processes: first-passage properties and connections with resetting". *Physical Review E* 114 (2), L022107
- 2) Huili Wang*, Junjie Xia*, Peer Mohamed Suhail Mujibur Rahman, Bijentimala Keisham, Abinash Padhi, Yiyu Deng, Yan Li, Luke Vistain, Sarah Kim, **Gabriel Mercado-Vasquez**, Aly A. Khan, Marcus R. Clark, and Savaş Tay. (2026) "Spatial Proximity Sequencing Maps Developmental Dynamics in the Germinal Center". *Cell*, 2026; 0
- 3) **Gabriel Mercado-Vasquez**, James V. Vizzard, Betul Celiker, Junjie Xia and Savaş Tay. (In revision) "Accurate processing of ultra-short immune signals by single macrophages".
- 4) Andrew G Wang*, **Gabriel Mercado-Vasquez***, Minjun Son*, Emma Kenna, James V Vizzard, Abinash Padhi, Huili Wang, Bijentimala Keisham, Aleksandr Gorin, Mustafa Bilgic, Alexander Hoffmann, and Savaş Tay. (In revision) "Encoding innate immune memory in single macrophages through cell intrinsic and paracrine feedback".
- 5) Junjie Xia, Bijentimala Keisham, Jia Meng Pang, Abinash Padhi, Huili Wang, **Gabriel Mercado-Vasquez**, Mia L. Huang, Savaş Tay, (In revision) "Integrated profiling of proteins, glycoforms and mRNA in single cells reveals dynamic glycan remodeling during T cell differentiation"
- 6) Yoon Jeong, **Gabriel Mercado-Vasquez**, Abinash Padhi, Yiyu Deng, Madhumitha Prakash, Jonathan M. Matthews, Sefika Ozcan, Erdal Toprak, and Savaş Tay. (In revision) "Massively multiplexed microfluidics maps combinatorial and sequential antibiotic responses in 3D".
- 7) Denis Boyer, **Gabriel Mercado-Vasquez**, Satya N. Majumdar and Grégory Schehr. (2024) "Optimizing the random search of a finite-lived target by a Lévy flight ". *Phys. Rev. E* 109, L022103
- 8) **Gabriel Mercado-Vasquez**, Denis Boyer, and Satya N. Majumdar. (2022) "Reducing mean first passage times with intermittent confining potentials: a realization of resetting processes". *Journal of Statistical Mechanics: Theory and Experiment*, vol. 2022.
- 9) **Gabriel Mercado-Vasquez**, Denis Boyer, and Satya N. Majumdar. (2022) "Freezing transitions of Brownian particles in confining potentials." *Journal of Statistical Mechanics: Theory and Experiment*, vol. 2022.
- 10) **Gabriel Mercado-Vasquez**, and Denis Boyer. (2021) "Search of stochastically gated targets with diffusive particles under resetting." *Journal of Physics A: Mathematical and Theoretical* 54.44.

- 11) **Gabriel Mercado-Vasquez** and Denis Boyer. (2021) "First hitting times between a run-and-tumble particle and a stochastically gated target." *Physical Review E*, vol. 103, p. 042139.
- 12) **Gabriel Mercado-Vasquez**, Denis Boyer, Satya N. Majumdar and Grégory Schehr. (2020) "Intermittent resetting potentials." *Journal of Statistical Mechanics: Theory and Experiment*, vol. 2020.
- 13) **Gabriel Mercado-Vasquez** and Denis Boyer. (2019) "First hitting times to intermittent targets." *Physical Review Letters*, vol. 123, p. 250603.
- 14) **Gabriel Mercado-Vasquez** and Denis Boyer. (2018) "Lotka–Volterra systems with stochastic resetting." *Journal of Physics A: Mathematical and Theoretical* 51.40: 405601.

GRANTS AND FELLOWSHIPS

• MRSEC grant: DMR-2011854. Outreach program	2026
• MRSEC grant: DMR-2011854. Outreach program	2025
• Outreach program "Clubes de Ciencia", funded by the University of Chicago	2024
• Ciencia de Frontera 2019 CONACYT, grant 10782	2019
• PhD. Scholarship CONACYT	2019
• DGAPA-PAPIIT IN108318 "Study of natural languages, genetic and protein"	2018
• Master scholarship CONACYT	2016
• DGAPA-PAPIIT IN111515 "Meso-structured materials on photo-controlled liberation of molecules and drugs".	2015
• DGAPA-PAPIIT IN111413 "Sol-gel nano-materials on non-linear optical catalysis of solar cells".}	2013

HONORS AND AWARDS

Alfonso Caso Medal to the most distinguished doctoral student per program National Autonomous University of Mexico, Mexico City, Mexico.	2026
Juan Manuel Lozano Mejía Medal for the best PhD thesis in Physics National Autonomous University of Mexico, Institute of Physics, Mexico City, Mexico.	2022
PhD degree with honors National Autonomous University of Mexico, Institute of Physics, Mexico City, Mexico.	2022
IOP Publishing "Outstanding Reviewer" Award Journal of Physics Communications.	2020
Juan Manuel Lozano Mejia Medal for the best master thesis in Physics National Autonomous University of Mexico, Institute of Physics, Mexico City, Mexico.	2019
Best poster award	2019

Mexican School of Statistical Physics and Complex Systems, Mexico City, Mexico.

Highlight publication

2018

Journal of Physics A: Mathematical and Theoretical, in the context of Biological Modelling.

PRESENTATIONS

"Cel-to-cell communication: information transmission in the immune response". Annual Symposium on Biomedicine and Immunology 2026, Mexico.	2026
"From Cells to Organs with AI" Symposium. National Institute for Theory and Math in Biology. Chicago, IL.	2026
"Information processing and immune response prediction via AI" (Poster). Science & Engineering Industry Expo 2025. Chicago, IL.	2025
"Automated system for study of cellular communication in hydrogel-based dynamic microenvironments". (Talk). Yoon Jeong, Gabriel Mercado, Abinash Padhi, Yiyu Deng, Madhumita Prakash, Savas Tay. MicroTAS, Montreal, CA.	2024
"Biological systems in intermittent environments: from stochastic searches to immune response" (Talk). Institute of Physics, UNAM. Mexico City, Mexico.	2024
"Limits of sensing in macrophages". Chicago BioEngineering Conference (Poster). Chicago, IL.	2024
"Intermittent resetting potentials" (Talk). The Beg Rohu Summer School 2022: Out of Equilibrium Dynamics. Beg Rohu, France.	2022
"Intermittent resetting potentials" (Poster). The Beg Rohu Summer School 2022: Out of Equilibrium Dynamics. Beg Rohu, France.	2022
"Random Search of Intermittent Targets" (Talk). XVII Congress of the Physical Science Graduate Students. Mexico City, Mexico.	2019
"First hitting time to intermittent targets" (Poster). Mexican School of Statistical Physics and Complex Systems, Mexico City, Mexico.	2019
"First hitting time to intermittent targets" (Poster). 2019 Arnold Sommerfeld School "on the Physics of Life", Munich, Germany.	2019
"Light induces flows in colloidal suspensions" (Poster). The LXI Mexican National Society of Physics Congress. Puebla, Mexico.	2018
"Lotka--Volterra Model with stochastic resetting" (Talk). Congress of the Physical Science Graduate Students. Mexico City, Mexico.	2017

“Experimental study of two annular vortex impact” (Poster). The LX Mexican National Society of Physics Congress. Monterrey, Mexico.	2017
“Experimental device to measure the angle of a linearly polarized light” (Poster). The LVII Mexican National Society of Physics Congress. Mazatlan, Mexico.	2014
“Low-cost robotic arm for medical applications” (Poster). The LVII Mexican National Society of Physics Congress. Mazatlan, Mexico.	2014

TEACHING AND MENTORING EXPERIENCE

The University of Chicago , Pritzker School of Molecular Engineering,	2022-Present
- Mentor for a PhD student’s Development Assistantship in the University of Chicago Biological Sciences Division (2026), guiding an independent, project-based professionalization term and providing formal evaluation of the student’s progress. - Mentored two postdoctoral researchers, four graduate students, and one high school student in technical training related to microfluidics, cell culture, optical microscopy, mathematical modeling, and simulations - Supervised experimental design and research projects for five mentees - Advised two graduate students through their PhD candidacy processes	
Directed the PhD research of one graduate student, providing guidance on research direction, milestones, and thesis development	
National Autonomous University of Mexico , Mexico City, Mexico	
<i>Teaching Assistant</i>	
Introduction to complex systems	2021
Statistical physics	2019-2021, 2017-2018
Electromagnetism II	2019
Electrical circuits and signals	2017
Digital machines with laboratory	2016
Robotics and controls	2016
Electromagnetism I	2014-2016

ACADEMIC SERVICE

Co-coordinator of the outreach program “Clubes de Ciencia”, funded by the University of Chicago	2026
Co-coordinator of the outreach program “Clubes de Ciencia”, funded by the University of Chicago	2025
Mathematical and Computational Applications; MDPI, Reviewer. 1 review	2024
Journal of Physics A: Mathematical and Theoretical, Reviewer. 1 review	2022
Physical Review Letters, Reviewer. 1 review	2021
Journal of Physics Communications, Reviewer. 1 review	2021
Physical Review E, Reviewer. 1 review	2020

OUTREACH AND ENGAGEMENT

Clubes de Ciencia México, Oaxaca, Mexico.	2026
South Side Science Festival, University of Chicago. Chicago, IL.	2025

- Clubes de Ciencia México, Oaxaca, Mexico. 2025
- South Side Science Festival, University of Chicago. Chicago, IL. 2024
- Biobits: Information Propagation in Nature (Workshop). Clubes de Ciencia México, Oaxaca, Mexico. 2024
- South Side Science Festival, University of Chicago. Chicago, IL. 2023

WORKSHOPS AND COURSES

- The Beg Rohu Summer School: Out of Equilibrium Dynamics. Saint Pierre Quiberon, France. 2022
- Mexican School of Statistical Physics and Complex Systems, Mexico City, Mexico. 2019
- Arnold Sommerfeld School "on the Physics of Life". Munich, Germany. 2019

TECHNICAL SKILLS

- Experimental setups: digital controls, combining microfluidics and bulk optics, microfluidics fabrication
- Analytics: simulations of stochastic dynamics, Gillespie algorithm, inverse Laplace transforms, finite element methods
- Programming languages: C++, Java, Fortran 90, Python, ASM, Matlab, Mathematica, LabView
- Software Programs: Windows, Linux, LaTeX, Word, Excel, Origin, PowerPoint, Beamer
- Cell culture: mammalian cell culture and bacterial culture
- Molecular biology: PCR reactions, cloning,
- Imaging: fluorescent microscopy

REFERENCES

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